MIL-D-60031A (AR)
19 May 1978
SUPERSEDING
MIL-D-60031 (MU)
15 April 1964

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M69
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of initiating device designated as Detonator, Electric, M69.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions
MIL-P-14077 - Polytetraflouroethylene (Teflon), Molding and Extruding Material

STANDARDS

MILITARY

MIL-STD-105D - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331 - Fuze and Fuze Components Environmental and performance tests for

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-883 - Test Methods and Procedures for Microelectronics

DRAWINGS

ARMY

8857199 - Detonator Electric, M69
8798698 - Box, Set-up Packing, Ammunition for Detonators Electric
8798699 - Box, Packing, Ammunition for Detonators Electric
8800784 - Ammunition Container Metal for Detonator Electric
9327302 - Transportation Vibration Fixture Assembly for M69 Electric Detonator
9327307 - Mechanical Shock Test Fixture Assembly for M69 Electric Detonator

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator.-The detonator shall comply with all requirements specified on Drawing 8857199, and with all requirements specified in applicable specifications.

3.3 Moisture content of PETN.-The moisture content of the PETN used in loading the detonator, at the time of loading, shall not exceed 0.50 percent.

3.4 Moisture content of lead azide.-The moisture content of the lead azide used in loading the detonator, at the time of loading, shall not exceed 0.50 percent.

3.5 Waterproofness.-The detonators shall be waterproof as evidenced by normal functioning after being immersed in water, maintained at a temperature of 20 degrees Centigrade plus or minus one (1) degree Centigrade for not less than 48 hours at a depth of 2 to 3 inches.

3.6 Environmental tests

3.6.1 Transportation vibration.-The detonator shall withstand the transportation vibration test as evidenced by satisfactory functioning after being subjected to the transportation vibration test specified in MIL-STD-331, Test No. 104, Procedure II.

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3.6.2 Mechanical shock.-The detonator shall withstand the mechanical shock test, as evidenced by satisfactory functioning, after being subjected to the mechanical shock test specified in Standard MIL-STD-883, Method 2002.2, test condition G, except that the shock pulse of half sine waveform shall be applied once only, for a minimum of 0.1 milliseconds duration, with the direction of acceleration perpendicular to the longitudinal axis of the detonator.

3.7 Functioning

3.7.1 Time.-The detonator shall fire on the first attempt within five (5) microseconds, when the electrical energy from a .0022 microfarad capacitor, maximum (max.), charged to 300 volts max., is applied to the detonator.

3.7.2 Output.-The detonator shall fire on the first attempt when the electrical energy from a .0022 microfarad capacitor max., charged to 300 volts max., is applied to the detonator and the detonator shall cause perforation of the lead disc to a diameter of not less than 3/16 inch.

3.7.3 Duds.-The detonator shall fire on the first attempt when the electrical energy from a .0022 microfarad capacitor max., charged to 300 volts, max., is applied to the detonator.

3.8 Workmanship.-All parts shall be fabricated and finished in a thorough workmanlike manner. The parts shall be free of burrs, chips, cracks, sharp edges, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

3.9 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are part of the detailed specification.

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4.2 Classification of inspections.-The following types of inspections shall be conducted on this item:

a. First Article Inspection of detonators.
b. Production Quality Conformance Inspection of detonators

4.3 First article inspection.-These tests are to be conducted to assure the quality of the detonators prior to the start of regular production.

4.3.1 Submission.-The manufacturer of the detonators, prior to the initiation of regular production, shall ship to a Government approved facility designated by the Contracting Officer for evaluation, an initial production sample consisting of one hundred and eighty (180) completed detonators and twenty five (25) bridge and plug assemblies for evaluation in accordance with the provisions of 4.3.2.

4.3.2 Inspections to be performed.-The first article sample assemblies shall be inspected for conformance with any or all of the requirements of this specification and any or all of the requirements of the applicable drawings (See Table I).

4.3.3 Rejection.-(See MIL-A-48078 (PA)).

4.3.4 Process and process controls.-The contractor, prior to the initiation of production shall submit for approval to the Contracting Officer, a written plan of processes and process controls for the following areas considered to be critical to functioning and reliability:

a. Weight control of all explosives used in loading the detonators (notes 4 and 6 of dwg 8857199).
b. Consolidation pressure controls of all explosive charges (note 5 and 7 of dwg 8857199).

The original plan and any changes to the process or process controls shall require approval by US Army Armament Research and Development Command, ATTN: DRDAR-LCR-E, Dover NJ 07801.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

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b. Lead azide from one batch or blending of batches from one manufacturer.

c. PETN from one batch or blending of batches from one manufacturer.

4.4.2 Examination.-One hundred percent examination shall be performed for all critical defects and major defects when so specified. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plans of Standard MIL-STD-105, Level II. All non-conforming material shall be rejected (see MIL-A-48078).

NOTE: Critical defect reporting.-A report of all critical defects found, number of parts or assemblies inspected for critical defects, and corrective action taken, if any, shall be compiled monthly for the duration of the contract and copies forwarded to the Contracting Officer and Commanding General, US Army Armament Research and Development Command, ATTN: DRDAR-LCR-E and DRDAR-QA-A-F, Dover, New Jersey 07801.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Detonator, Electric, M69		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bridge and Plug Assembly Dwg. 8857197 Examination of defects	25	(a)	3.2/3.8	4.4.2.1
	Detonator, Electric, M69 Dwg. 8857199 Examination of defects		(a)	3.2/3.8	4.4.2.2
	Waterproofness and output	60	(a)	3.5/3.7.2	4.5.4
	Functioning time	60	(a)	3.7.1	4.5.5
	Transportation vibration and duds	30	(a)	3.6.1	4.5.6.1
	Mechanical shock and duds	30	(a)	3.6.2	4.5.6.2

NOTES: a. To be performed at contractor's plant.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Bridge and Plug Assembly		8857197
			NEXT HIGHER ASSEMBLY 8857199

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101.	Lacquer on bridge.		0.40%	3.2	Visual
102.	Spot charge missing, cracked, porous, or separated.		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship.		0.65%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Detonator, Electric, M69		8857199
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Cup cracked.		100%	3.2	Visual
2.	Explosive on exterior.		100%	3.2	Visual
3.	Resistance between pin and cup.		100%	3.2	Test (4.5.3)
Major					
101.	Total length, max.		0.40%	3.2	Gage
102.	Outside diameter.		0.40%	3.2	Gage
103.	End of pin not clean.		0.40%	3.2	Visual
104.	Crimp not full 360 degrees.		0.40%	3.2	Visual
105.	Priming charge missing.		0.40%	3.2	Visual
106.	Base charge missing.		0.40%	3.2	Visual
107.	Moisture seal missing or incomplete.		0.40%	3.2	Visual
Minor					
201.	Evidence of poor workmanship.		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8800784
4.4.2.3	Ammunition, Container, Metal				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Contents exposed.		0.40%	3.2/3.8	Visual
102.	Sealing strip incomplete or wrinkled.		0.40%	3.2/3.8	Visual
Minor					
201.	Contents loose.		0.65%	3.2/3.8	Visual
202.	Marking incorrect, incomplete or illegible.		0.65%	3.2/3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Box, Packing, Set-up		8790698
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101.	Waterproof seal broken.		0.40%	3.2/3.8	Visual
102.	Sealing strip improper.		0.40%	3.2/3.8	Visual
Minor					
201.	Contents loose.		0.65%	3.2/3.8	Visual
202.	Marking incorrect, incomplete or illegible.		0.65%	3.2/3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Sealed Barrier Bag				8793693
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101.	Seal improper or incomplete.		0.40%	3.2/3.8	Visual
102.	Bag punctured, torn or cut.		0.40%	3.2/3.8	Visual
Minor					
201.	Marking missing, misleading or unidentifiable.		0.65%	3.2/3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60031A (MR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Box, Packing				6798699
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101.	Strapping, missing, broken or loose.		0.40%	3.2/3.6	Visual-Manual
102.	Contents exposed.		0.40%	3.2/3.6	Visual
Minor					
201.	Contents loose.		0.65%	3.2/3.6	Manual
202.	Marking misleading or illegible.		0.65%	3.2/3.6	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Moisture content of PETN (see 3.3).-The moisture content of the PETN shall be determined for each lot at the time of loading. A sample of 5 grams shall be selected from each batch and tested as specified in 4.5.1. If the moisture content is in excess of the percentage specified and loading has not begun, the batch of PETN represented by the sample shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.5.1. If loading has begun, action on the unloaded PETN shall be as specified above and all detonators loaded with the questionable PETN shall be rejected.

4.4.3.2 Moisture content of lead azide (see 3.4):-The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected from each batch and tested as specified in 4.5.2. If the moisture content is in excess of the percentage specified and loading has not begun, the batch of lead azide represented by the sample shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.5.2. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.4.3.3 Waterproofness (and output), transportation vibration, mechanical shock and functioning.-A sample of one hundred and eighty (180) detonators shall be selected from each lot and the sample apportioned as specified below for testing in the sequence specified.

4.4.3.3.1 Waterproofness (and output) (see 3.5 and 3.7.2) Sixty (60) of the sample detonators selected in 4.4.3.3 above shall be subjected to this test. The test shall be performed as specified in 4.5.4, using test equipment specified in 4.4.4. If one or more detonators fail to produce a perforation in the disc of the required size the lot shall be rejected. Detonators that fail to fire shall be recorded and replaced with sample detonators from the lot.

4.4.3.3.2 Functioning time (see 3.7.1).-Sixty (60) detonators of the sample selected in 4.4.3.3 above shall be subjected to this test. The test shall be performed as specified in 4.5.5, using test equipment specified in 4.4.4. The lot shall be rejected if the mean functioning time (x) plus the product of 2.27 times the standard deviation of the sample (s) exceeds 5 microseconds. Detonators that fail to fire shall not be

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considered in the required calculation, but will be recorded and replaced with sample detonators from the lot. The standard deviation shall be calculated using the formula 6.5 or its equivalent. If two (2) or more detonators fail to fire in the first and second test combined the lot shall be rejected and further testing stopped.

4.4.3.3.3 Transportation vibration (see 3.6.1 and 4.5.6.1) mechanical shock (see 3.6.2 and 4.5.6.2) and duds (see 3.7.3). The remaining samples selected in 4.4.3.3 above shall be subjected to the following tests in the sequence specified.

4.4.3.3.3.1 Transportation vibration.-Thirty (30) of the detonators selected in 4.4.3.3 above shall be subjected to this test.

Defect classification is shown in the following:

Critical Defect	Major Defect
1. Breakage of, damage to or loosening of any part likely to result in a safety hazard during test or handling after test.	1. Failure to fire (dud) after TV Test.
2. Detonator functioning during TV test.	

If one or more "critical defects" occur the lot shall be immediately rejected and the finding reported. If only one dud occurs a second sample of thirty (30) detonators shall be selected, conditioned and tested. If one (1) more dud occurs in the second sample the lot shall be rejected.

4.4.3.3.3.2 Mechanical shock.-The remaining thirty (30) sample detonators selected in 4.4.3.3 above shall be subjected to the mechanical shock test specified in Standard MIL-STD-883, Method 2002.2. Defect classification shall be as follows:

Critical Defects	Major Defect
1. Breakage of, damage to or loosening of any part likely to result in a safety hazard during test or handling after test.	1. Failure to fire (dud) after mechanical shock test.
2. Detonator functioning during mechanical shock test.	

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If one or more "Critical" defects occur the lot shall be immediately rejected and the finding reported. If only one dud occurs a second sample of thirty (30) detonators shall be selected, conditioned and tested. If one more dud occurs in the second sample the lot shall be rejected.

4.4.3.4 Heat seal test of sealed bag (see dwg. 8798698) Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examination and tests prescribed herein is described in the Paragraph/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedure

4.5.1 Determination of moisture content of PETN.-Approximately 5 grams of PETN shall be placed in a tared glass dish. The dish and contents shall be weighed, dried at 95 degrees Centigrade (°C), plus or minus 2 degrees C, for one hour, cooled in a desiccator and weighed. The loss in weight shall be calculated to the percentage of moisture in the PETN.

4.5.2 Determination of moisture content of lead azide A tared dried aluminum dish, 2 inches in diameter shall be covered with a sheet of teflon (see MIL-P-14077), 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two (2) grams shall be transferred to the moisture dish while the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95 degrees C, for 2 hours. The cover shall be placed in a sulfuric acid dessicator during the drying period. After drying, the dish and contents shall be quickly transferred to the dessicator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to the percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{A \text{ minus } B \times 100}{\text{Divided by } A.}$$

where:
A = the weight of the sample before drying.
B = the weight of the sample after drying.

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4.5.3 Resistance of pin and cup.-The resistance of pin and cup shall be measured using an electrical ohmmeter, using not more than 20 microamperes of current. Any detonator exhibiting resistance outside the specified drawing limits shall be classed defective.

4.5.4 Waterproofness and output.-The detonator shall be conditioned as specified in 3.5. The detonator shall be removed from conditioning and assembled to the applicable test fixture and within one (1) hour fired by releasing the energy from a .0022 microfarad capacitor, max., charged to 300 volts max. Observation shall be made for perforation of the lead disc to the required size and for duds.

4.5.5 Functioning time.-The detonator shall be assembled to the applicable test fixture and connected electrically to the specified test circuit. The energy from a .0022 microfarad capacitor, max., charged to 300 volts, max., shall be discharged to fire the detonator. The functioning time shall be measured as the time from application of energy until functioning of the detonator causes the electronic counter to stop. The time shall be recorded for Standard Deviation calculation and appropriate action taken with respect to duds.

4.5.6 Transportation vibration at minus 40 Degrees Fahrenheit (°F), +70°F., and plus 125°F., mechanical shock at minus 40°F., plus 70°F., and plus 125°F., and duds

4.5.6.1 Transportation vibration and testing for duds at extreme temperatures.- The sample detonators shall be tested in accordance with the procedures specified in Standard MIL-STD-331, using the test fixture specified on dwg. 9327302 for compliance with the requirements. After successful compliance with the requirements, the tested detonators shall be divided into three (3) equal groups. One (1) group shall then be conditioned at minus 40°F for 2 hours minimum, removed from the conditioning chamber and fired within 3 minutes, max. The second group shall be conditioned at plus 125 degrees F. for 2 hours minimum, removed from the conditioning chamber and fired within 3 minutes, max. The third group shall be fired at ambient temperature. The detonators shall be fired by discharging the electrical energy from a .0022 microfarad capacitor, max., charged to 300 volts, max. Any detonator that fails to fire shall be classed defective.

4.5.6.2 Mechanical shock and testing for duds at extreme temperatures.-The sample detonators shall be tested in accordance with the procedures specified in Standard MIL-STD-883, using the test fixture specified on dwg. 9327307 for compliance

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with the requirements. After successful compliance with the requirements, the tested detonators shall be divided into three (3) equal groups. One (1) group shall then be conditioned at minus 40°F. for 2 hours, minimum, removed from the conditioning chamber and fired within 3 minutes, max. The second group shall be conditioned at plus 125°F. for 2 hours, min., removed from the conditioning chamber and fired within 3 minutes, max. The third group shall be fired at ambient temperature. The detonators shall be fired by discharging the electrical energy from a .0022 microfarad capacitor, max., charged to 300 volts, max. Any detonator that fails to fire shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C - minimum military package.-The detonators shall be preserved and packaged in accordance with dwg. 8800784.

5.1.2 Intermediate packaging.-The detonators shall be packaged in accordance with dwg. 8798698.

5.2 Packing

5.2.1 Level C - minimum military pack.-The containers shall be packed in accordance with dwg. 8798699.

5.3 Marking

5.3.1 Unit.-Intermediate and exterior containers shall be marked in accordance with dwgs. 8798698, 8798699 and 8800784.

6. NOTES

6.1 Intended use.-The detonator covered by this specification is intended for use in Fuze, Point Initiating, Base Detonating, M509A1E1 and Fuze, Point Initiating, Base Detonating, M530A1.

6.2 Ordering data.-(See 6.1 of MIL-A-48078 (PA).

6.3 Submission of design approval.-See 6.2 of MIL-A-48078. Submit equipment designs, as required, to Commander, ARRADCOM HQ, Dover, NJ 07801, ATTN: DRDAR-QA-A-F.

6.4 Check test for possible deterioration.-The following test is designed to prevent the loading of detonators which have deteriorated into a major ammunition item. The test

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should be performed on detonators from each lot which have been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require the check to be performed, the Contracting Officer should insert a provision for the performance of the check test in the contract.

6.4.1 Waterproofness and output, transportation vibration mechanical shock, functioning time and duds.-A sample of two hundred and forty (240) detonators shall be selected from each lot and the sample apportioned as specified below for testing in the sequence specified.

6.4.1.1 Waterproofness and output.-Sixty (60) of the sample detonators selected in 6.4.1 above shall be subjected to this test. The test shall be performed as specified in 4.5.4, using test equipment specified in 4.4.4. If two (2) or more detonators fail to produce a perforation in the disc of the required size the lot shall be rejected. Detonators that fail to fire shall be replaced with detonators from the lot.

6.4.1.2 Functioning time.-Sixty (60) detonators of the sample detonators selected in 6.4.1 above shall be subjected to the functioning time test specified in 4.5.5, using test equipment specified in 4.4.4. The lot shall be rejected if the mean functioning time (X) plus the product of 2.25 times the standard deviation of the same (s) exceeds 5 microseconds. Detonators that fail to fire shall not be considered in the required calculation but will be replaced with detonators from the lot, however, if four (4) or more detonators fail to fire in the first and second test combined the lot shall be rejected and further testing stopped.

6.4.1.3 Transportation vibration, mechanical shock and duds.- If the lot has not been rejected for the tests in 6.4.1.1 and 6.4.1.2 above, the remaining samples selected in 6.4.1 shall be subjected to the following tests in the sequence specified.

6.4.1.3.1 Transportation vibration and duds.-Sixty detonators of the sample selected in 6.4.1 shall be tested as specified in 4.5.6.1, using test equipment specified in 4.4.4. If two (2) or more detonators fail in any phase the lot shall be rejected.

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6.4.1.3.2 Mechanical shock and dud.-Sixty detonators of the sample detonators selected in 6.4.1 shall be tested as specified in 4.5.6.2, using test equipment specified in 4.4.4 If two (2) or more detonators fail in any phase the lot shall be rejected.

6.5 Calculation of sample standard deviation.-The following formula or equivalent shall be used in calculating the sample standard deviation:

$$S = \text{sum of } (x_i - \bar{x})^2 \text{ divided by } N - 1$$

where:
S = sample standard deviation
X = numerical value of individual observations
$\bar{X}$ = average value of the sample
N - sample size

6.6 Check test expense.-The Contracting Officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

Custodian:
Army - (AR)

Preparing Activity:
Army - (AR)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-D-60031A Detonator, electric, M69 LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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